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Research on spray cooling performance based on battery thermal management

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Summary
To balance the temperature uniformity and the compactness of the battery module, the heat transfer performance of spray cooling technology is theoretically analyzed in this article. The main factors affecting the heat transfer performance of spray cooling are discussed. Through the experiments, the influence of spray cooling on the thermal management of the battery thermal management system under high discharge rate is studied. The results show that, in spray-cooling 4 + 2.5 m/s cooling mode, the total heat transfer coefficient K reaches 201.0 W/(m^2 K), which is 409.3% higher than the heat transfer coefficient of the forced air cooling. In addition, the difference between the spray concentration of the heat source surface and the spray concentration of the mainstream area during steady-state heat dissipation is an important factor affecting the heat transfer performance of spray cooling. Moreover, spray cooling can provide a lower temperature rise and a more uniform temperature distribution for the battery module. For the spray-cooling 4 + 4 m/s case, the maximum temperature rise is only 10.3°C, and the temperature difference among the battery module is stably maintained within 5°C. Therefore, the proposed spray cooling battery thermal management system shows to be efficient approach in handling the heat accumulation of battery module.

KEYWORDS
heat transfer performance, power battery, spray cooling, thermal management

1 | INTRODUCTION

Under the dual pressure of energy crisis and environmental problems, countries all over the world have begun to formulate energy development strategies and develop new and renewable energy sources for the purpose of energy conservation and environmental protection.[1-3]

The development of electric vehicles (EV) and hybrid electric vehicles (HEV) is considered to be one of the effective measures to alleviate the increasing pressure of environmental and energy crisis.[4-6] With the rapid development of electric vehicles, the demand for high energy density power batteries has also grown rapidly. With the increase in battery energy density and the expansion of applications, the degradation of vehicle performance and safety problems caused by poor battery thermal management have become increasingly prominent. Specifically, the main reasons for the thermal safety problems of

Abbreviations: EV, electric vehicles; HEV, hybrid electric vehicles; PCM, phase change material; spray-cooling $i + j$ m/s, spray power i + air cooling (air flow rate j m/s).

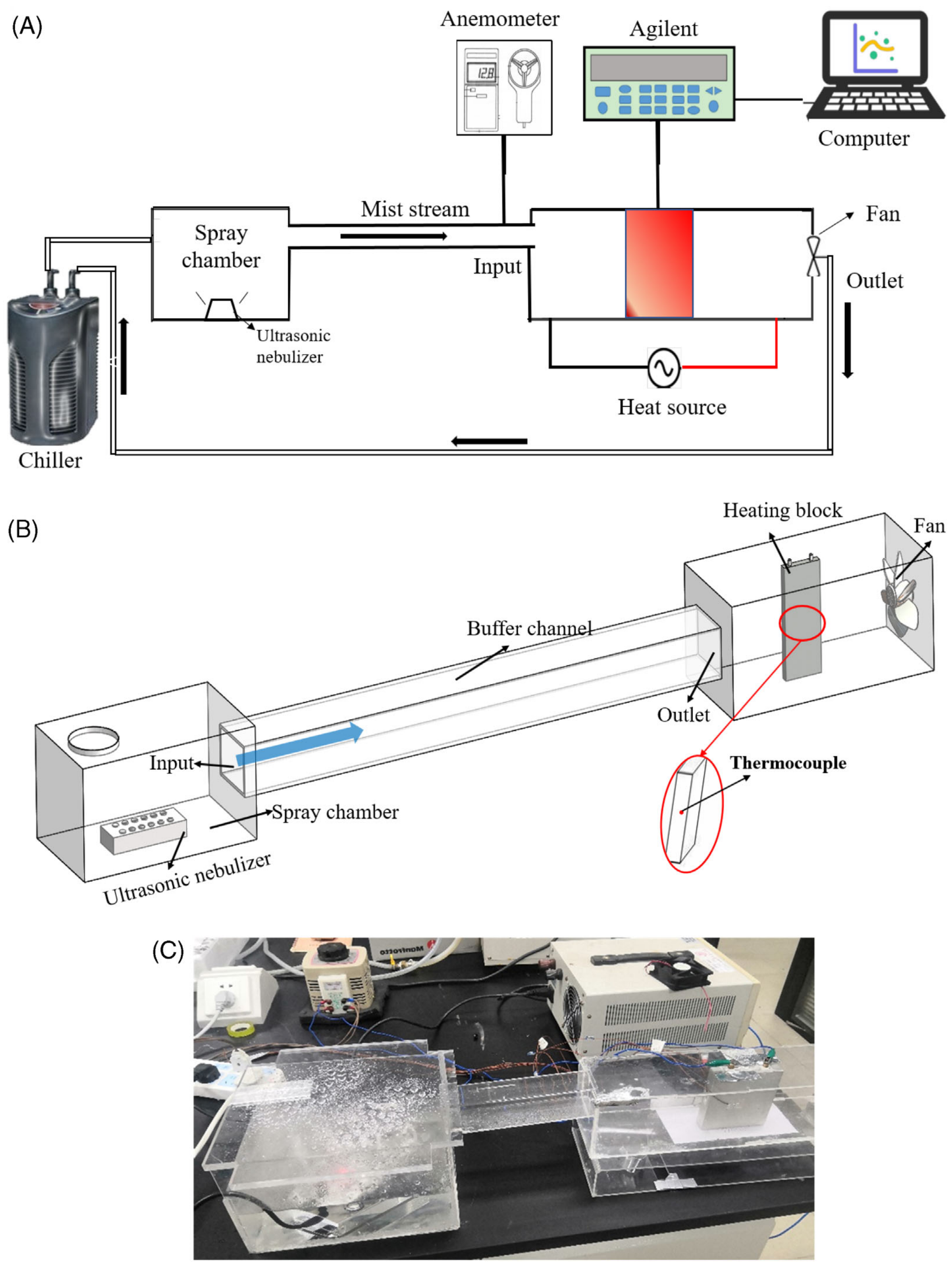

FIGURE 1 The spray cooling system (A) The schematic diagram of the spray cooling, (B) the spray cooling module, and (C) photo of spray cooling module

power batteries include: (a) During high rate or multiple charge/discharge cycles, the internal electrochemical reaction causes the battery temperature to rise sharply, and the phenomenon follows the chain reaction mechanism. The decomposition reaction of battery component materials will occur one after another, and then thermal runaway will occur.[7] (b) The increase in the charging rate and the serious local heat flow will cause uneven heating of the single battery, which will cause irreversible damage, degradation of battery module performance and safety accidents.[8] Therefore, under the current bottleneck of battery technology, the development of an effective battery thermal management system is an effective way to solve the thermal problem of the battery system.

Related research pointed out that the ideal operating temperature range of lithium-ion batteries is 25°C to 50°C,[9] and the maximum temperature difference should be less than 5°C in the entire battery pack.[10] In this regard, to dissipate the heat of lithium-ion battery packs, numerous research has been conducted on active and passive thermal management methods, such as air cooling,[10,11] liquid cooling,[11] and phase change cooling.[12] The air cooling system has advantage of simple structure, low cost and lightness, but it has the problems of low heat capacity and low thermal conductivity.[13,14] Although liquid-cooled systems are more efficient at dissipating heat than air-cooled systems, they require additional complex liquid piping and connecting components, adding volume, weight, and maintenance costs to the overall system.[15] Phase change materials can store and release heat in the process of phase change, thereby controlling the temperature of the battery.[16] However, under high rate discharge or repeated extreme cycle conditions, the PCM reaches thermal saturation and cannot release the accumulated heat in time, which will lead to thermal failure.[17] It is worth mentioning that each cooling system has its own advantages and disadvantages, and no thermal management system alone has an absolute advantage over others. The choice among air cooling, liquid cooling and PCM cooling systems depends entirely on the specific combination of temperature, heat rate, battery capacity, and geometry.

Of all the thermal management systems available today, air cooling is the first choice.[18] This is mainly due to its lower operating costs, compact design, and good reliability. Therefore, how to balance the compactness and temperature uniformity of the battery module while keeping the air cooling structure simple is a new challenge at present. This prompted researchers to find a

TABLE 1 Atomization amount under different spray power numbers

Spray power	Atomization amount (g/h)
Spray-cooling 1	176.0
Spray-cooling 2	296.1
Spray-cooling 3	529.2
Spray-cooling 4	1664.9
Spray-cooling 5	2418.8

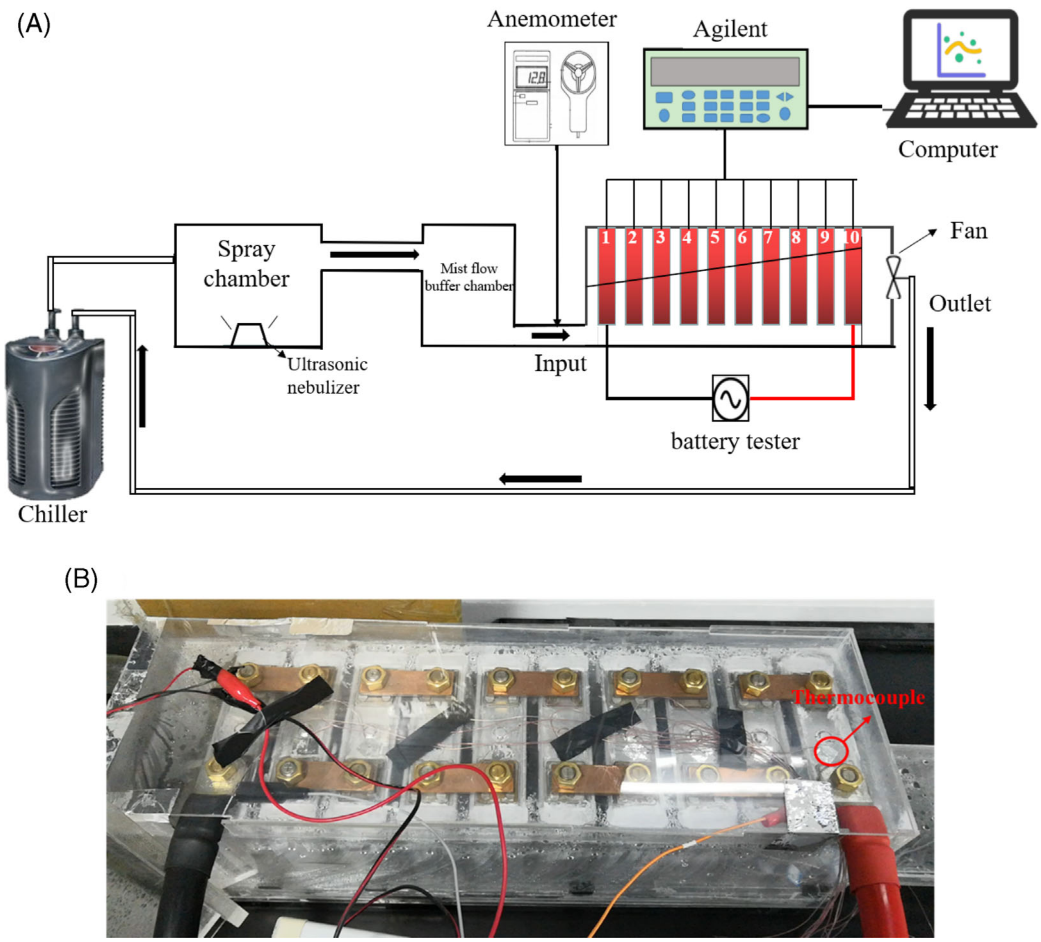

FIGURE 2 The experimental test platform of the spray cooling system. (A) Schematic diagram of spray cooling battery thermal management system, (B) photo of battery module

more efficient and economical new battery thermal management method. Evaporative cooling becomes an emerging cooling technology used in current thermal management systems, since it can dissipate high heat flow at a reasonable cost.[19,20] Compared with the single-phase forced convection process, the evaporative cooling of the phase change process from liquid to gas has superior heat transfer performance.[21] In addition, the phase change from liquid to gas has greater latent heat than the phase change from solid to liquid. The heat of water evaporation is about 2400 kJ/kg, which is 9 to 11 times that of paraffin melting.[21] Therefore, since the evaporative cooling technology was proposed, due to its excellent performance, it has been widely used in the fields of building energy saving,[22] industrial large-scale equipment cooling,[23] and electronic chip cooling.[24] In recent years, this technology has been introduced into battery thermal management systems. Lei et al[25] combined phase change materials with sprays, which effectively suppressed the increase in battery temperature and reduced the temperature difference in the battery module at high temperatures. Lip HuatSaw et al[18] proposed the use of spray cooling technology in the battery pack thermal management system, and studied the heat dissipation performance of conventional air cooling method and spray cooling method through numerical simulations and experiments. The results showed that, compared with dry air cooling, spray cooling can provide a more uniform temperature distribution. And the surface temperature of the battery module can be kept below 40°C. The above research results show that spray cooling has an excellent effect in enhancing heat transfer. However, in the field of power battery thermal management, the literature on spray cooling is still relatively rare.

In summary, the spray cooling technology is an effective way to reduce the temperature rise of the battery caused by the excessively high local heat flow density, and to control and realize the uniform temperature distribution in the battery module. However, there are few reports on the quantitative analysis of spray cooling effect and the process of heat and mass transfer on the surface of the heat source during spray cooling. In addition, under different working conditions, the changing law of the limit heat transfer coefficient of spray cooling, and how to balance the relationship between simple structure and battery thermal management performance is very crucial. Therefore, in this work, the heat transfer performance of spray cooling technology was theoretically analyzed. The main factors that affected the heat transfer performance of spray cooling were explored, and different experimental studies were conducted to find out the influence of the factors. In order to further verify the feasibility of spray cooling in the field of battery thermal management, a power battery thermal management system using spray cooling was built, and the effect of high discharge rate on the thermal management of the battery thermal management system was studied.

2 | EXPERIMENTAL DESIGN

2.1 | Spray cooling heat transfer performance test system

In order to investigate the heat and mass transfer laws in the spray cooling process under different working conditions and provide guidance for the design of the spray cooling power battery thermal management system, the spray cooling performance test system shown in Figure 1 is designed. As shown in Figure 1A, the power of the ultrasonic atomizer is controlled through the controller. The flow speed is adjusted by controlling the DC power supply, and measured with an anemometer. The heating power of the heat source is controlled by adjusting the AC power supply. Risheng small chiller MINI-200 is used in condenser. The temperature of the heat source is collected by a T-type thermocouple with an accuracy of ±1°C. The thermocouple is placed in a hole pre-made in the center of the heat source. The signal measured by the thermocouple is processed by Agilent 34972A and recorded by the computer. In this experiment, a custom aluminum block with a length of 130 mm, a width of 36 mm, and a height of 108 mm was employed as the heat source. The spray cooling module is shown in Figure 1B,C. When the heat generation and heat dissipation of the heat source reach a balance, that is, the temperature of the heat source remains stable, the data collection is stopped. The spray is atomized in the spray chamber and driven by a fan, and it flows through the surface of the heat source through the buffer channel to cool the heat source. Table 1 lists the spray concentration under different spray power.

TABLE 2 Properties of the battery used in this study

Parameter	Value
Max operation temperature during charge (°C)	0-45
Max operation temperature during discharge (°C)	−20 to 60
Optimum operation temperature during charge (°C)	15-35
Optimum operation temperature during discharge (°C)	15-35
End voltage (V)	2.0
Charging voltage (V)	3.65
Nominal voltage (V)	3.2

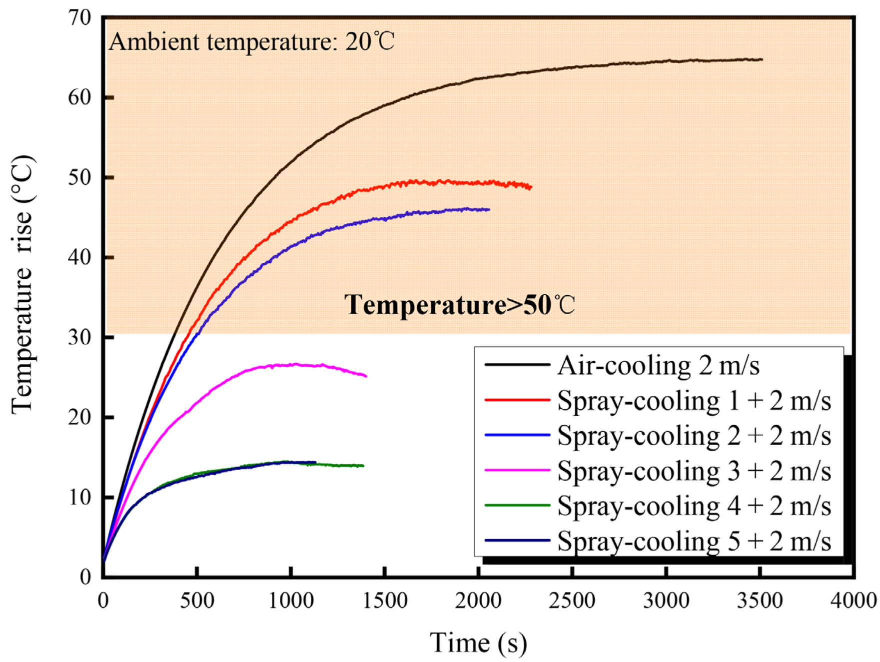

FIGURE 3 Heat source temperature rise under different spray power

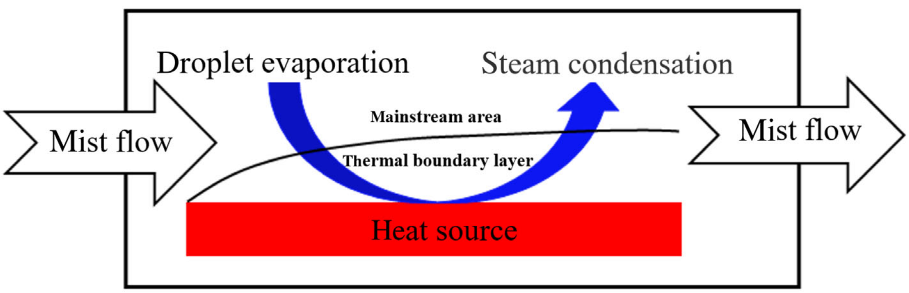

FIGURE 4 Schematic diagram of heat transfer mechanism

The total uncertainty of the experimental results is mainly caused by the inaccuracy of power input, thermocouple, wind speed tester and ultrasonic atomizer. Assuming that y is a function of the independent variables x_1, x_2, ..., x_n, y can be calculated by Equation (1).[26] Therefore, the total uncertainty of the experiment is calculated as 4.58% according to the following formula.

$$\delta y = \left[\left(\frac{\partial y}{\partial x_1}\delta x_1\right)^2 + \left(\frac{\partial y}{\partial x_2}\delta x_2\right)^2 + \cdots + \left(\frac{\partial y}{\partial x_n}\delta x_n\right)^2\right]^{\frac{1}{2}} \quad (1)$$

2.2 | The performance of the spray cooling system

In order to further verify the feasibility of spray cooling in the field of battery thermal management, a spray cooling battery thermal management system was designed. Ten single cells are connected in series to form a battery module. During the test, the discharge rate is 5C (C-rate is the ratio of the charge/discharge current of the battery to the nominal capacity), and the ambient temperature is 30°C. The spray cooling system is composed of an atomization module and an air cooling module. Figure 2 shows the experimental test platform for the thermal management performance of the spray cooling system. The cooling fan provides the power of the spray flow, and the anemometer (accuracy of $\pm 3\% \pm 0.1$ dgts) measures the inlet flow speed. After the atomizer is started, the water mist and the inlet air are mixed in the spray chamber to form a spray flow. The spray flows through the buffer chamber and flows into the battery module chamber. After the heat transfer process in the battery module chamber, the spray forms condensed water and flows back to the spray chamber. The surface temperature of the battery module is recorded by thermocouples placed near the poles. In this experiment, a commercial prismatic lithium battery with a length of 130 mm, width of 36 mm, and a height of 108 mm (provided by Chunlan Clean Energy Research Institute Co., Ltd.) is employed. The battery parameters are listed in Table 2.

3 | RESULTS AND DISCUSSION

3.1 | The heat transfer mechanism of spray cooling

The temperature rise of the heat source at different spray powers is shown in Figure 3. The smaller the temperature rises on the surface of the heat source which indicates better heat transfer performance. It can be seen from Figure 3 that, when forced air cooling is used, the highest temperature on the surface of the heat source rises to 64.7°C. While using spray cooling, the temperature rise of the heat source is significantly reduced. In addition, as the spray power increases, the temperature rise of the heat source gradually decreases. The enhanced heat transfer process is shown in Figure 4. Compared with forced air cooling, the heat transfer enhancement effects of spray flow on the heat source mainly come from the following aspects: (a) When the spray flows in the main stream zone, the droplet group absorbs the sensible heat of the air and evaporates, reducing the temperature of the air in the main flow area. (b) The evaporation of droplets in the spray flow in the thermal boundary layer changes the temperature distribution gradient in the thermal boundary layer, resulting in a change in the thickness of the thermal boundary layer. (c) The droplet group in the spray flow directly exchanges heat when it touches the heat source, and after absorbing a large amount of heat from the heat source, it is brought into the mainstream area through the mass transfer process.

Furthermore, it can be seen in Figure 3, as the spray power increases, the heating rate of the heat source, the highest temperature rise and the time required to reach the steady state are gradually reduced. During the cooling modes of spray-cooling $1 + 2$ m/s and spray-cooling $2 + 2$ m/s, the temperature of the heat source drops significantly, but it is still higher than 50°C. This is mainly because the evaporation rate of the spray flow is higher than its replenishment rate. Further increasing the spray power, the heat source temperature can be controlled below 50°C. This is mainly due to the increase in the spray power which leads to an increase in the concentration of the spray flow and effectively enhances the heat transfer capability of the spray cooling. It is worth noting that in the case of spray-cooling 4 and spray-cooling 5, as the spray power continues to increase, the temperature rise on the heat source surface basically does not change. This should be due to the fact that when the concentration of water mist increases to a certain value, the evaporation rate on the heat source is lower than the replenishment rate of the droplet group, which limits the further increase of its heat exchange capacity.

3.2 | The heat transfer coefficient of spray cooling

In order to quantitatively analyze the heat transfer effect of spray cooling, the actual heat transfer coefficient (K) and the limit heat transfer coefficient (K_m) are derived. The actual physical process of the heat transfer of spray cooling on the surface of the heat source is very complicated. In order to simplify the calculation, the flow phenomenon involved in spray cooling is simplified to a two-dimensional, steady-state incompressible fluid sweeping flat plate flow. Some assumptions are made as follows:

1. The volumetric force of the fluid is ignored heat dissipation, chemical reaction heat, and energy transfer caused by molecular diffusion are ignored.
2. The influence of water mist on the fluid flow field is ignored.
3. The spray flow evaporates only when it touches the surface of the heat source.

For the flow on the heat source surface, the mass and heat transfer coefficient h_m is:

$$h_m = h\frac{D}{\lambda}\left(\frac{\mathrm{Sc}}{\mathrm{Pr}}\right)^{-\frac{2}{3}} = \frac{h}{\rho c_p}\mathrm{Le}^{-\frac{2}{3}} \tag{2}$$

where ρ is the air density, c_p is the specific heat capacity of the air at constant pressure, h_m is the mass transfer coefficient, h is the convection heat transfer coefficient, D is the diffusion coefficient, Sc is the Schmidt criterion number, Le is the Lewis criterion number, Pr is the Prandtl criterion number, and λ is the thermal conductivity of air.

During normal operation of the battery, its temperature is usually lower than 50°C. For the heat and moisture exchange occurring in the air, Le is usually regarded as a constant within the temperature range, and its value is about 1. In the case of ignoring radiation heat transfer, the heat transfer of spray cooling is mainly composed of the latent heat of droplet evaporation and the heat convective of the flow. According to the law of energy conservation and Equation (2), the heat flux q and the total heat transfer coefficient K can be expressed as:

$$q = h_m(\rho_H - \rho_0)Q + h(T - T_0) = \frac{Qh}{\rho c_p}(\rho_H - \rho_0) + h(T - T_0) \tag{3}$$

$$K = \frac{Qh}{\rho c_p (T - T_0)}(\rho_H - \rho_0) + h = \frac{Qh}{c_p(T - T_0)}(d_H - d_0) + h \tag{4}$$

TABLE 3 The heat transfer coefficient at different spray power

	Air-cooling	Spray-cooling 1	Spray-cooling 2	Spray-cooling 3	Spray-cooling 4	Spray-cooling 5
Flow speed (m/s)	2.0	2.0	2.0	2.0	2.0	2.0
Spray concentration (g/m^3)	0.0	9.8	16.5	29.4	92.5	134.4
Heating power (W)	51	51	51	51	51	51
K (W/(m^2 K))	32.7	42.6	45.8	81.3	150.6	151.1
K_m (W/(m^2 K))	—	147.6	147.6	147.6	147.6	147.6

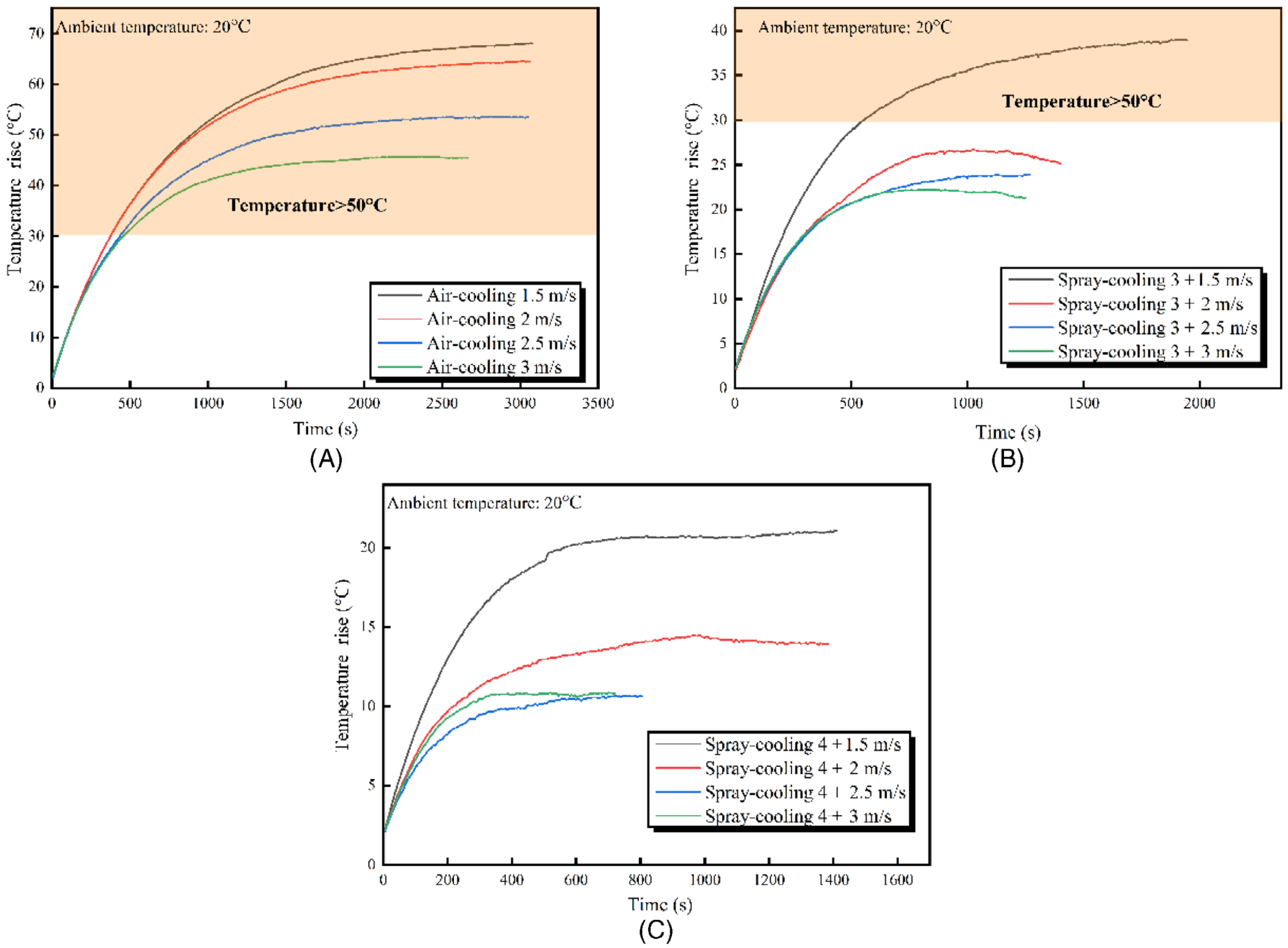

FIGURE 5 The effect of flow speed on the temperature rise. (A) The temperature rise of the heat source under air cooling mode. (B) The temperature rise of the heat source under spray-cooling 3 mode. (C) The temperature rise of heat source under spray-cooling 4 mode

where ρ_H is the absolute humidity of the heat source surface, ρ_0 is the absolute humidity at the main stream zone, T is the temperature of the heat source surface, T_0 is the temperature of the main stream zone, Q is the latent heat of vaporization of water, d_H is the moisture content of the heat source surface, and d_0 is the absolute moisture content of the mainstream zone.

Under standard atmospheric pressure, the moisture content in moist air can be calculated according to Equation (5)[27]:

$$d = 0.622\frac{\phi p_s}{p-\phi p_s} = 0.622\left(\frac{\frac{2}{15}\phi e^{18.5916-\frac{3991.11}{T-39.31}}}{101.3-\frac{2}{15}\phi e^{18.5916-\frac{3991.11}{T-39.31}}}\right) \quad (5)$$

where φ is relative humidity, p is atmospheric pressure, and p_s is the partial pressure of water vapor.

When the mass concentration of the spray flow on the hot surface reaches a saturated state, it is equivalent to covering a static water film on the hot surface, and convective evaporation heat transfer occurs. Then, the total heat transfer coefficient K_m is:

$$K_m = \frac{Qh}{c_p(T - T_0)}(d_T - \varphi d_0) + h \tag{6}$$

where d_T is the moisture content of saturated humid air at temperature T. Then the relationship between the steady-state temperature of spray cooling and the steady-state temperature of forced air cooling is:

$$\frac{Q}{c_p}(d_T - \phi d_0) + T_S = T_A \tag{7}$$

where T_A is the steady-state temperature of forced air cooling and T_S is the steady-state temperature of spray cooling.

It is known from the above analysis that the difference between the spray concentration of the heat source surface and the spray concentration of the mainstream area during steady-state heat dissipation is an important factor affecting the heat transfer performance of spray cooling. As shown in Table 3, the actual heat transfer coefficient (K) and the limit heat transfer coefficient (K_m) under different cooling modes are shown. It can be seen that the spray cooling method can effectively improve the heat transfer performance of the cooling system. As the spray power increases, the heat transfer coefficient gradually increases. When the spray power is lower than spray-cooling 4, due to the rapid evaporation of the spray flow, the liquid film cannot form, and the K value is less than K_m. However, under the cooling modes of spray-cooling 4 + 2 m/s and spray-cooling 5 + 2 m/s, the heat transfer coefficient maintains 150.6 and 151.1 W/(m^2 K), respectively. This indicates that, under the current heating power, the spray concentration on the heat source surface has reached saturation, which limits the further increase of the spray cooling heat transfer capacity. In addition, under the cooling modes of spray-cooling 4 + 2 m/s and spray-cooling 5 + 2 m/s, the value of K_m is slightly smaller than the value of K. This should be caused by the model assumption that the influence of flow speed on the evaporation rate of spray flow has been ignored. Based on the above analysis, it can be found that the spray-cooling 4 + 2 m/s mode should be the optimal cooling method. At this case, the evaporation rate and replenishment rate of the spray flow are basically conserved, and the best heat transfer effect is exhibited.

3.3 | The effect of flow speed

Figure 5 shows the effect of flow speed on the temperature rise. It can be seen from Figure 5A that, for the pure forced air cooling mode, the highest temperature of the heat source gradually decreases as the flow speed increases, and the time required to reach the steady state temperature gradually decline. When the flow speed is 3 m/s, the maximum temperature rise of the heat source surface and the time to reach the steady state are 45°C and 2677 seconds, respectively. As shown in Figure 5B,C, it can be seen that after the spray cooling is used, due to the huge latent heat of water and the enhancement of convection heat transfer, in the case of spray-cooling 3 + 1.5 m/s, the time required to reach the steady state is

TABLE 4 The effect of flow speed on the heat transfer coefficient

Performance parameter	Flow speed (m/s) 1.5	2.0	2.5	3.0
Air-cooling				
Maximum temperature rise (°C)	68.1	64.7	53.6	45.5
K (W/(m^2 K))	30.9	32.7	39.5	46.4
Spray-cooling 3				
Spray concentration (g/m^3)	39.2	29.4	23.5	19.6
K (W/(m^2 K))	54.3	82.2	88.5	100.0
K_m (W/(m^2 K))	142.6	147.6	170.0	193.2
Spray-cooling 4				
Spray concentration (g/m^3)	123.3	92.5	74.0	61.7
K (W/(m^2 K))	101.6	150.6	201.0	195.9
K_m (W/(m^2 K))	142.6	147.6	170.0	193.2

only 2014 seconds, and the maximum temperature rise is 39°C. With the increase of flow speed, the heat transfer process shows an increasing trend, but the increasing rate shows a downward trend. It can be seen form Figure 5C, the cooling mode of spray-cooling 4 + 3 m/s takes less time to reach thermal equilibrium, while the cooling mode of spray-cooling 4 + 2.5 m/s exhibits lower temperature rise. According to the Equation (3), the heat transfer of spray cooling is mainly composed of the latent heat of droplet evaporation and the heat convective of the flow. When the flow rate is increased, the convective heat transfer performance is improved, which will also lead to a decrease in the spray concentration, thereby weakening the heat transfer performance of the droplet evaporation. Since the both cooling modes exhibit comparable cooling performance, it can be inferred that it is difficult to improve the cooling performance by continuing to increase the flow speed. The cooling mode of spray-cooling 4 + 2.5 m/s should be the best cooling condition.

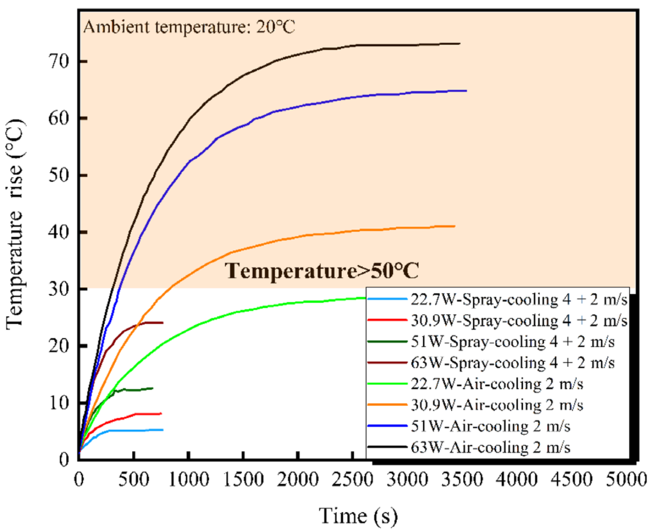

FIGURE 6 Heat source temperature rise under different heating powers

Table 4 shows the influence of different flow speeds on the heat transfer coefficient. It can be seen that in the case of spray-cooling 3 + 3 m/s, the heat transfer coefficient of spray cooling reaches a maximum of 100 W/(m^2 K). Compared with forced air cooling at the same flow speed, the heat transfer coefficient is enhanced by 115.4%. It is worth mentioning that in spray-cooling 3 mode, the heat transfer performance is significantly improved with the increase of flow speed, but it does not reach the limit heat transfer coefficient (K_m). This is because the increase in flow speed not only enhances the convective heat transfer, but also leads to a decrease in the spray concentration on the heat source, which indicates that the surface of the heat source cannot form a liquid film due to the low spray concentration. As for the spray-cooling 4 + 2.5 m/s mode, the heat transfer coefficient reaches 201.0 W/(m^2 K). Compared with forced air cooling mode, the heat transfer coefficient is enhanced by 409.3%. Moreover, further increasing the flow speed to 3 m/s, the heat transfer performance will decrease. It is caused by the decrease in the spray concentration on the heat source, which indicates weaker evaporative heat transfer.

3.4 | The effect of heating power

Different heating power represent different working conditions of the battery. Figure 6 shows the effect of different heating power on the temperature rise of the heat source. It can be seen that, as the power of the heat source increases, the heating speed of the heat source and the highest temperature rise to the steady state increase. After adopting the spray cooling, compared with air cooling, the cooling performance of the system is greatly improved. At the heating power of 63 W, the maximum temperature rise dropped from 74.0°C to 24.2°C.

The specific influence of heating power on spray cooling is shown in Table 5. It can be seen that when

TABLE 5 The influence of different heating powers on spray cooling

	Heating power (W)			
Performance parameter	**22.7**	**30.9**	**51.0**	**63.0**
Flow speed (m/s)	2.0	2.0	2.0	2.0
Air cooling				
Maximum temperature rise (°C)	28.5	41.0	64.8	74.0
K (W/(m^2 K))	32.2	31.0	32.7	35.1
Spray-cooling 4				
Maximum temperature rise (°C)	6.6	9.0	14.5	24.2
K (W/(m^2 K))	142.6	140.2	150.6	110.0
K_m (W/(m^2 K))	127.8	127.5	147.6	165.1

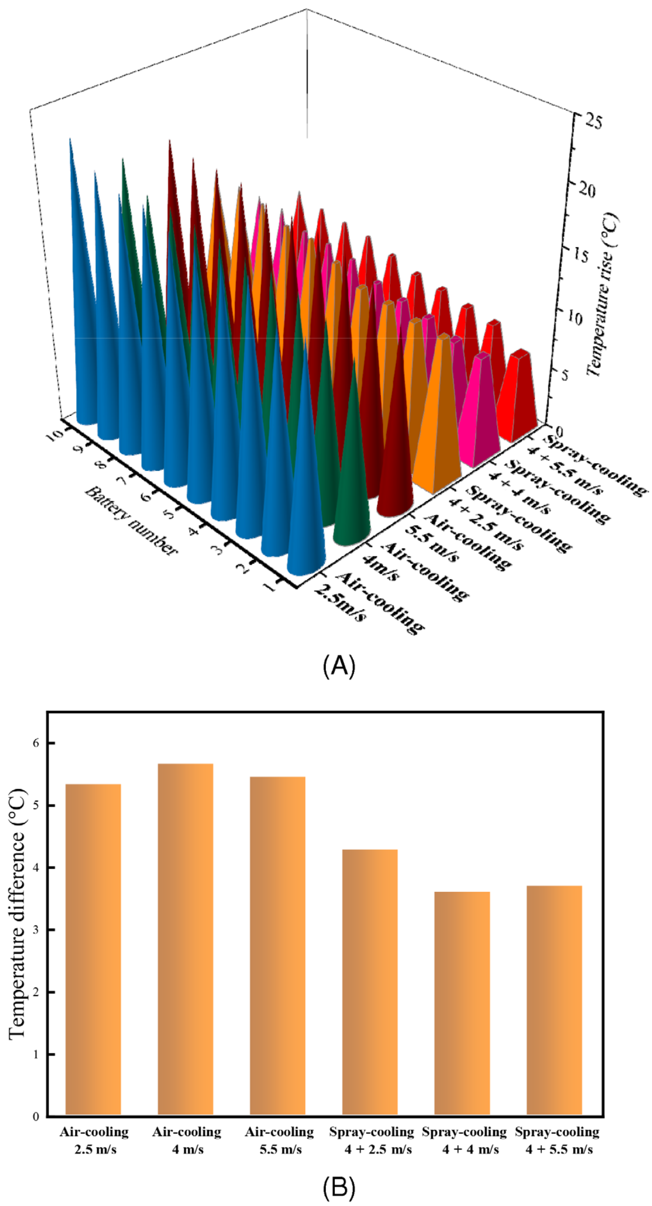

FIGURE 7 Temperature distribution in the battery module. (A) Maximum temperature rise on the surface of the battery. (B) Temperature difference among the battery module

spray cooling is used, with the heating power less than 51 W, the maximum temperature rise and the heat transfer coefficient slightly increase with the heating power. When the heating power is 63 W, the heat transfer coefficient of spray cooling is significantly reduced. The reason should be that when the spray flow concentration and flow speed are constant, the increase of heating power leads to the increase of the mass transfer rate of water droplets on the hot surface. When the evaporation rate of the droplets group on the hot surface is greater than the replenishment rate, the spray cooling heat transfer performance will be reduced.

3.5 | The effect of the spray cooling on the battery thermal management system

In order to further verify the effectiveness of the spray cooling method on the thermal management system, the temperature difference distribution and temperature rise of the battery under high-rate discharge were tested at an ambient temperature of 30°C. Figure 7 shows the temperature distribution of the battery module at a discharge rate of 5C. Figure 7A shows the maximum temperature rise and Figure 7B shows temperature difference among the battery module at the end of discharge. For the forced air cooling cases, when the flow speed is 2.5, 4.0, and 5.5 m/s, the maximum temperature rise of the battery module is 22.9°C, 20.2°C, and 20.4°C, respectively. Meanwhile, the temperature difference of the battery module exceeds 5°C. Both the battery temperature and the temperature difference have exceeded the safety threshold. As for the spray cooling cases, the temperature of the battery surface is greatly reduced at the same flow speed. It can be found that, when the flow speed is 2.5, 4.0, and 5.5 m/s, the spray cooling method can effectively control the temperature of the battery under a safe threshold, and its cooling performance is significantly improved with the increase of flow speed. For the spray-cooling 4 + 4 m/s case, the maximum temperature rise is only 10.3°C, and the temperature difference among the battery module is stably maintained within 5°C. Overall, the spray cooling method has a good thermal management effect on the battery module, and the good temperature consistency inside the module ensures that the battery module can still operate safely under high current conditions.

4 | CONCLUSIONS

In this work, the spray flow is used as the heat dissipation medium instead of air, and the heat transfer performance of the spray cooling technology is theoretically analyzed and experimentally studied. The proposed spray-cooled battery thermal management system is an effective method to deal with the heat storage of battery components and provides a potential solution for battery thermal management. The main conclusions are shown as follows:

1. In the spray cooling system, the difference between the spray concentration of the heat source surface and the spray concentration of the mainstream area during steady-state heat dissipation is an important factor affecting the heat transfer performance of spray cooling.
2. In the case of a heating power of 51 W, the spray cooling 4 + 2.5 m/s mode shows the best heat transfer

effect. In this mode, K reaches 201.0 W/(m^2 K), which is 409.3% higher than the heat transfer coefficient of forced air cooling.

3. With the constant spray concentration and flow rate, when the evaporation rate of the hot surface is greater than the replenishment rate of the droplet group, the heat transfer capacity of the system will decrease. It can be seen that for the cooling mode of spray-cooling $4 + 2$ m/s, when the heating power is increased from 51 to 63 W, the heat transfer coefficient of spray cooling is reduced from 150.6 to 110.0 W/(m^2 K), and the heat transfer performance significantly reduced.
4. Under extreme working conditions (5C discharge), the spray cooling method can reduce the temperature rise of the battery and achieve uniform temperature distribution in the battery module. In the case of spray-cooling $4 + 4$ m/s, the maximum temperature rise is only 10.3°C, and the temperature difference between battery modules is kept within 5°C.

It is worth mentioning that the current results are valuable for designing and optimizing more comprehensive thermal management systems in large-scale applications such as electric vehicles and electric energy storage. In addition, in future researches, the impact of different coolants and droplet sizes on spray cooling systems will be studied.

NOMENCLATURE

h_m	mass and heat transfer coefficient (m/s)
h	convection heat transfer coefficient (W/(m^2 °C))
D	diffusion coefficient (m^2/°C)
Sc	Schmidt criterion number
Le	Lewis criterion number
Pr	Prandtl criterion number
c_p	specific heat capacity (J/(kg °C))
ρ_H	absolute humidity of the heat source surface (kg/m^3)
ρ_0	absolute humidity at the main stream zone (kg/m^3)
T	temperature of the heat source surface (°C)
T_0	temperature of the main stream zone (°C)
Q	latent heat of vaporization of water (J/g)
d_H	moisture content of the heat source surface (kg/kg dry air)
d_0	absolute moisture content of the mainstream zone (kg/kg dry air)
d	moisture content (kg/kg dry air)
p	atmospheric pressure (Pa)
p_s	partial pressure of water vapor (Pa)
$\mathrm{K_m}$	total heat transfer coefficient (W/(m^2 °C))
d_T	moisture content of saturated humid air at temperature T (kg/kg dry air)
T_A	steady-state temperature of forced air cooling (°C)
T_S	steady-state temperature of spray cooling (°C)

GREEK SYMBOLS

λ	thermal conductivity (W/(m K))
ρ	density (kg/m^3)
φ	relative humidity (%)

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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